

Multiple Choice (10 marks)

1. To produce light, the atoms inside a light bulb transform electrical energy into what other kind of energy?
 - (a) heat
 - (b) chemical
 - (c) mechanical
 - (d) electromagnetic
2. Which of the following is the primary type of stress that a fixed vise exerts on a piece of wood?
 - (a) tensile
 - (b) shear
 - (c) torsion
 - (d) compression
3. Why is water from an aquifer more likely to be cleaner than water from other sources?
 - (a) because it forms where fresh and salt water meet
 - (b) because it receives water directly from precipitation
 - (c) because it rises to the surface near the ocean
 - (d) because pollutants are filtered by rock and soil deep within Earth
4. Which form of energy can travel through a vacuum?
 - (a) thermal
 - (b) solar
 - (c) mechanical
 - (d) chemical
5. What is the major process of surface rock formation on volcanoes?
 - (a) Rock cools quickly from melted rock.
 - (b) Rock is recrystallized by extreme pressure.
 - (c) Rock solidifies slowly deep underground.
 - (d) Rock is formed from deposited sediment.

True / False (10 marks)

Answer True or False

6. True or False: A toy truck will roll faster on a surface covered with sand than on a smooth surface.
7. True or False: Radiation from the Sun is the main energy source for deep ocean currents that circulate large volumes of water around the planet.
8. True or False: A thermal insulator has a low specific heat capacity, allowing it to absorb heat quickly.
9. True or False: Matter must always exist in a fixed shape and cannot change its form.
10. True or False: A book on a flat table moves horizontally because a balanced force is being exerted on it.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of chromosomes in human reproduction, explaining what occurs when male and female gametes unite to form offspring and how this impacts genetic variation.
12. Discuss how the drag of wind influences the characteristics and movement of ocean surface currents, providing examples to illustrate your points.

End of Paper